

TASK 2 : Deep Learning - Resume Classification System

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Internship Platform: InternSpark (Powered by Alfido Tech)

Task: Deep Learning - Resume Classification

1. Objective

Build a Deep Learning based Resume Classification System using NLP and a Bidirectional LSTM model to automatically classify resumes into job categories.

2. Project Overview

The project loads resume data, cleans text, tokenizes resumes, trains a Bidirectional LSTM model and evaluates performance using multiple metrics.

3. Dataset Information

Dataset: resume_dataset.csv

Rows: 166

Columns: Category, Resume

4. Technologies Used

Python

TensorFlow

Pandas

NumPy

Matplotlib

Scikit-learn

Jupyter Notebook

NLP

Bidirectional LSTM

5. Notebook Information

Notebook: train.ipynb

Model File: resume_classifier.keras

Generated Files: accuracy.png, loss.png

6. GitHub Code Link

Paste your GitHub repository link here after uploading the project.

7. Methodology

Load dataset → Clean text → Tokenize → Pad sequences → Encode labels → Build BiLSTM
→ Train → Evaluate model.

8. Results and Screenshots

Dataset Information

```
Shape: (166, 2)
Columns:
['Category', 'Resume']
<class 'pandas.DataFrame'>
Index: 166 entries, 0 to 168
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Category    166 non-null    str
1   Resume      166 non-null    str
dtypes: str(2)
memory usage: 3.9 KB
```

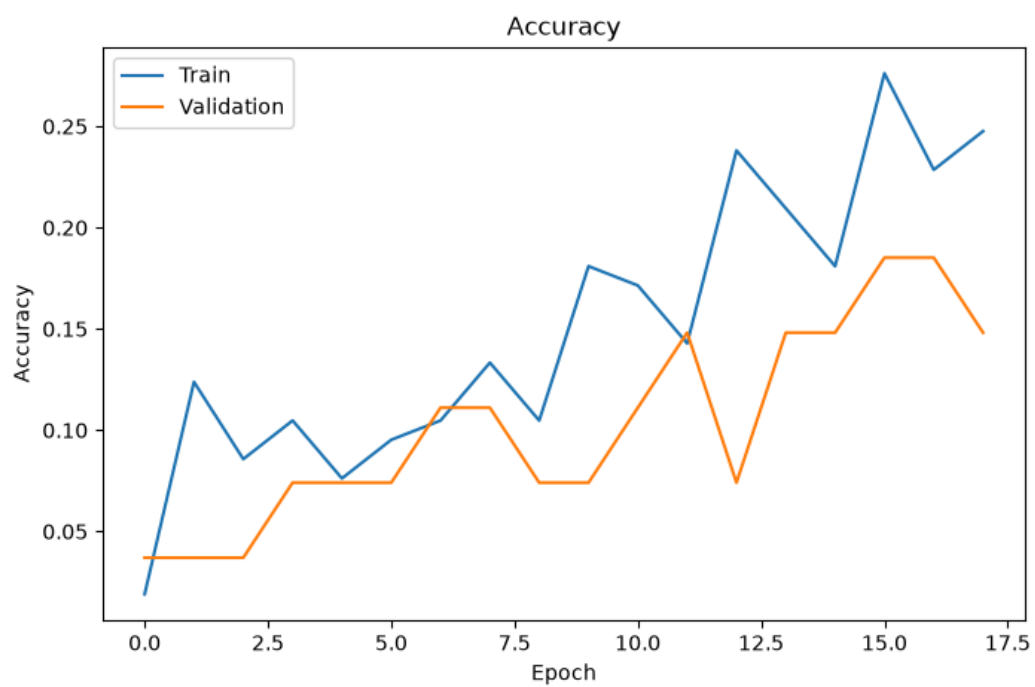
Explanation: Dataset Information output generated during model training and evaluation.

Training Progress

```
7/7 ██████████ 2s 246ms/step - accuracy: 0.1238 - loss: 3.2069 - val_accuracy: 0.0370 - val_loss: 3.2179
Epoch 3/20
7/7 ██████████ 2s 250ms/step - accuracy: 0.0857 - loss: 3.2005 - val_accuracy: 0.0370 - val_loss: 3.2129
Epoch 4/20
7/7 ██████████ 2s 247ms/step - accuracy: 0.1048 - loss: 3.1887 - val_accuracy: 0.0741 - val_loss: 3.2088
Epoch 5/20
7/7 ██████████ 2s 256ms/step - accuracy: 0.0762 - loss: 3.1669 - val_accuracy: 0.0741 - val_loss: 3.2025
Epoch 6/20
7/7 ██████████ 2s 260ms/step - accuracy: 0.0952 - loss: 3.1529 - val_accuracy: 0.0741 - val_loss: 3.1947
Epoch 7/20
7/7 ██████████ 2s 258ms/step - accuracy: 0.1048 - loss: 3.1108 - val_accuracy: 0.1111 - val_loss: 3.1813
Epoch 8/20
7/7 ██████████ 2s 247ms/step - accuracy: 0.1333 - loss: 3.0415 - val_accuracy: 0.1111 - val_loss: 3.1654
Epoch 9/20
7/7 ██████████ 2s 238ms/step - accuracy: 0.1048 - loss: 2.9650 - val_accuracy: 0.0741 - val_loss: 3.1260
Epoch 10/20
7/7 ██████████ 2s 272ms/step - accuracy: 0.1810 - loss: 2.9229 - val_accuracy: 0.0741 - val_loss: 3.1243
Epoch 11/20
7/7 ██████████ 2s 264ms/step - accuracy: 0.1714 - loss: 2.8675 - val_accuracy: 0.1111 - val_loss: 3.1198
Epoch 12/20
7/7 ██████████ 2s 281ms/step - accuracy: 0.1429 - loss: 2.8213 - val_accuracy: 0.1481 - val_loss: 3.1080
Epoch 13/20
...
Epoch 17/20
7/7 ██████████ 2s 281ms/step - accuracy: 0.2286 - loss: 2.5123 - val_accuracy: 0.1852 - val_loss: 3.0376
Epoch 18/20
7/7 ██████████ 2s 245ms/step - accuracy: 0.2476 - loss: 2.4878 - val accuracy: 0.1481 - val loss: 3.0471
```

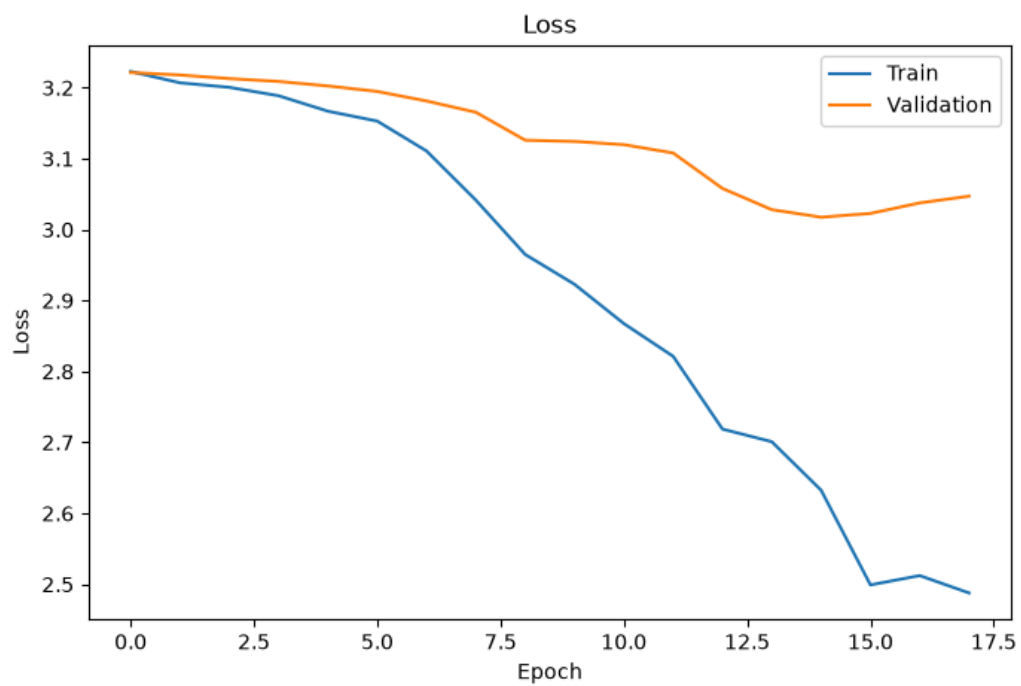
Explanation: Training Progress output generated during model training and evaluation.

Accuracy Graph



Explanation: Accuracy Graph output generated during model training and evaluation.

Loss Graph



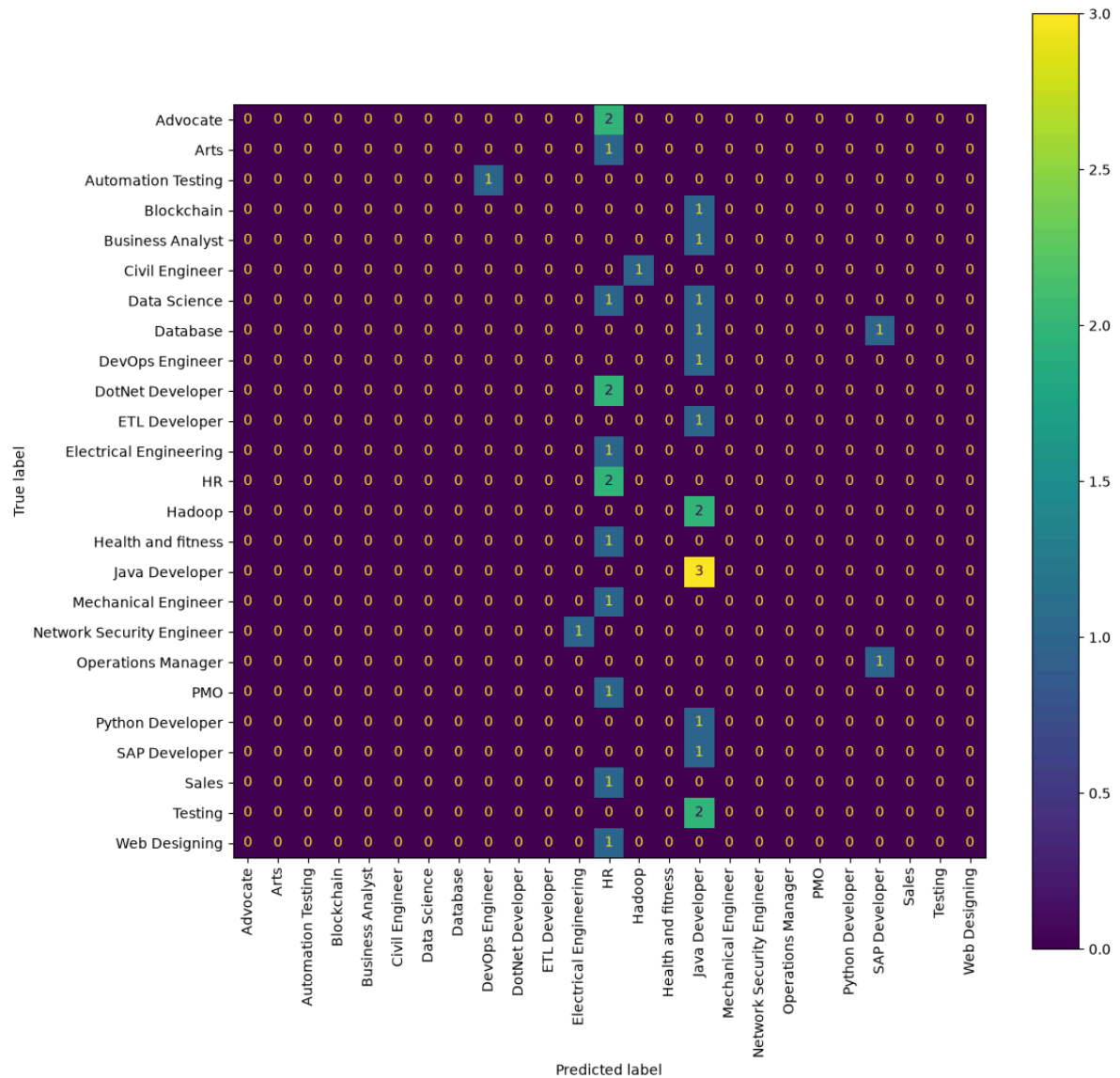
Explanation: Loss Graph output generated during model training and evaluation.

Classification Report

...	precision	recall	f1-score	support
Advocate	0.00	0.00	0.00	2
Arts	0.00	0.00	0.00	1
Automation Testing	0.00	0.00	0.00	1
Blockchain	0.00	0.00	0.00	1
Business Analyst	0.00	0.00	0.00	1
Civil Engineer	0.00	0.00	0.00	1
Data Science	0.00	0.00	0.00	2
Database	0.00	0.00	0.00	2
DevOps Engineer	0.00	0.00	0.00	1
DotNet Developer	0.00	0.00	0.00	2
ETL Developer	0.00	0.00	0.00	1
Electrical Engineering	0.00	0.00	0.00	1
HR	0.14	1.00	0.25	2
Hadoop	0.00	0.00	0.00	2
Health and fitness	0.00	0.00	0.00	1
Java Developer	0.20	1.00	0.33	3
Mechanical Engineer	0.00	0.00	0.00	1
Network Security Engineer	0.00	0.00	0.00	1
Operations Manager	0.00	0.00	0.00	1
PMO	0.00	0.00	0.00	1
Python Developer	0.00	0.00	0.00	1
SAP Developer	0.00	0.00	0.00	1
Sales	0.00	0.00	0.00	1
...				
accuracy			0.15	34
macro avg	0.01	0.08	0.02	34

Explanation: Classification Report output generated during model training and evaluation.

Confusion Matrix



Explanation: Confusion Matrix output generated during model training and evaluation.

9. Observations

The dataset is relatively small which affects model accuracy. Additional data and balancing categories can improve performance.

10. Conclusion

The Resume Classification System was successfully built and evaluated using Deep Learning techniques.

11. Future Improvements

Use larger datasets, balance classes, use BERT and deploy the model in production environments.